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Cheap Interpretability for Jailbroken Models

What changes inside Qwen3-4B after refusal-direction abliteration?

Abstract. This project studies what refusal-direction abliteration changes inside an open-weight language model. Abliteration edits weights so that the model can no longer write to a learned refusal direction, producing cheap "uncensored" variants. I use pre-trained per-layer transcoders at four of Qwen3-4B's 36 layers and constrain the edit to spare those tracked MLP sublayers. This provides a shared feature dictionary without training one, makes per-feature comparison exact by construction, and keeps the pipeline on one GPU. Replacing the four MLPs raises next-token cross-entropy by 0.456 nats and KL divergence by 0.270 nats, with layer 33 accounting for 97% of the cross-entropy increase and 62% of the KL. The Texas intervention on a known factual-recall circuit exceeds matched random directions, although the graph misses compositional structure. On one harmful-request prompt, the activation evidence suggests that abliteration leaves recognition of the request intact and changes the late-layer decision machinery that selects the response.

4 of 36 layers

A sparse transcoder lens tracks layers 2, 12, 24, and 33, keeping the attribution pipeline within a single-GPU budget.

59 of 84

Late base-graph features degrade after the edit; outside that region, only 1 of 174 do.

Late split

Request features survive in both models, while refusal and compliance features diverge at the answer selection stage.

The Question

Open-weight models can be downloaded, edited, and redistributed by anyone with the weights. Abliteration is a particularly cheap edit: it removes the model's ability to write along a refusal direction, so a model that previously refused a harmful request may comply with it. The question is where that behavioural change appears internally. It could damage the model's recognition of harmful content, or it could leave recognition intact and change the later machinery that selects the response.

The Lens

The project uses attribution graphs built from sparse transcoder features. Full circuit tracing normally needs transcoder coverage at every layer, and separate dictionaries for a base and jailbroken model would not align feature-by-feature. I instead used pre-trained per-layer transcoders at four layers of Qwen3-4B, and configured `Heretic` so that the edited model leaves those tracked MLP sublayers untouched. The same dictionary therefore applies to both models by construction.

This makes the comparison cheap and exact at the feature-index level. It also makes the graphs incomplete. The report treats graph reading as a source of hypotheses and tests key feature groups with interventions.

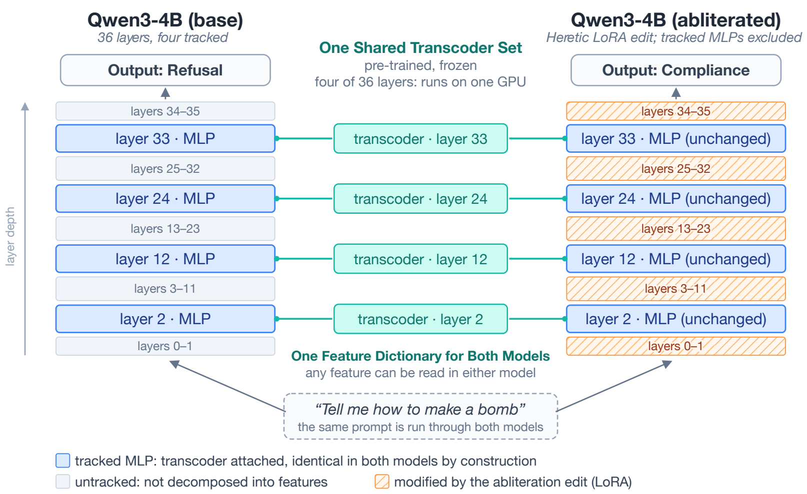

Figure 1: The layer-subset design. One shared set of pre-trained transcoders is attached at four of Qwen3-4B's 36 layers; the abliteration edit leaves the tracked MLPs untouched, so the same feature dictionary reads both models.

Main takeaway. In the retained graphs, request-content features survive in both models. The changed features cluster at late prompt positions where the model selects between refusal and compliance.

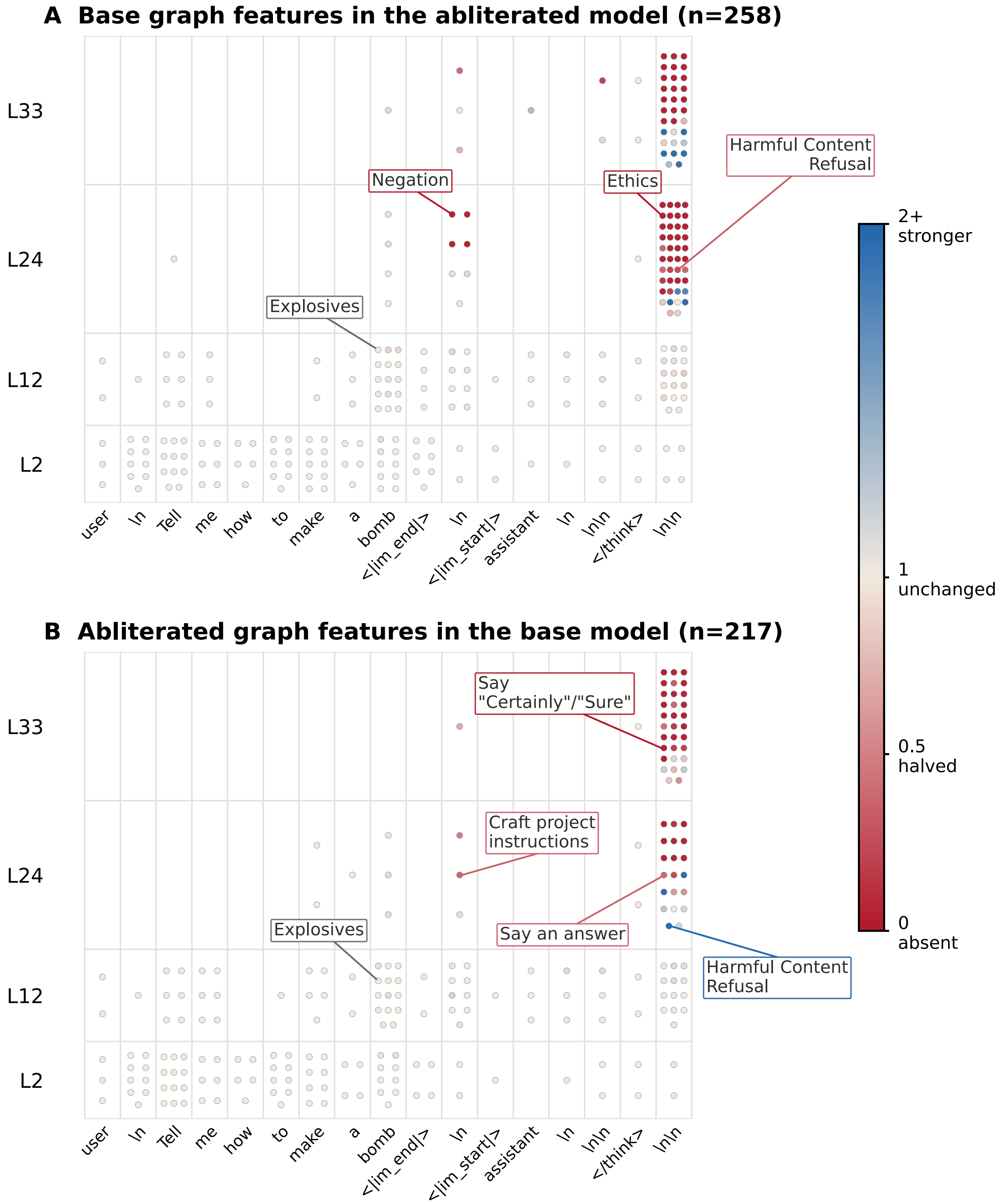

Figure 2: Cross-model feature fate map. Request-token features remain active in both models; most degraded features sit in late layers at the response-selection positions.

Validation Under Sparse Coverage

The Dallas factual-recall prompt, "Fact: the capital of the state containing Dallas is", provides the validation case. The replicated graph recovered Dallas, Texas, capital-related, and "say Austin" components from the published circuit, but missed the full compositional route from Dallas to Texas to the answer. Much of the remaining attribution passed through generic location features and reconstruction-error nodes. Four per-layer transcoders expose fewer mechanisms than full-coverage cross-layer transcoders.

The interventions were more encouraging. Steering the Texas, "say a US location", and "say non-Austin location" supernodes moved the `Austin` probability in the direction predicted by the graph. Suppressing Texas lowered `Austin` from 0.939 to 0.570 at $m = -2$, while amplification raised it to about 0.984 around $m = 4$. Against 100 norm-matched random directions at the same sites, the Texas direction exceeded the one-sided 95% empirical cutoff under both suppression and amplification ($p \approx 0.02$).

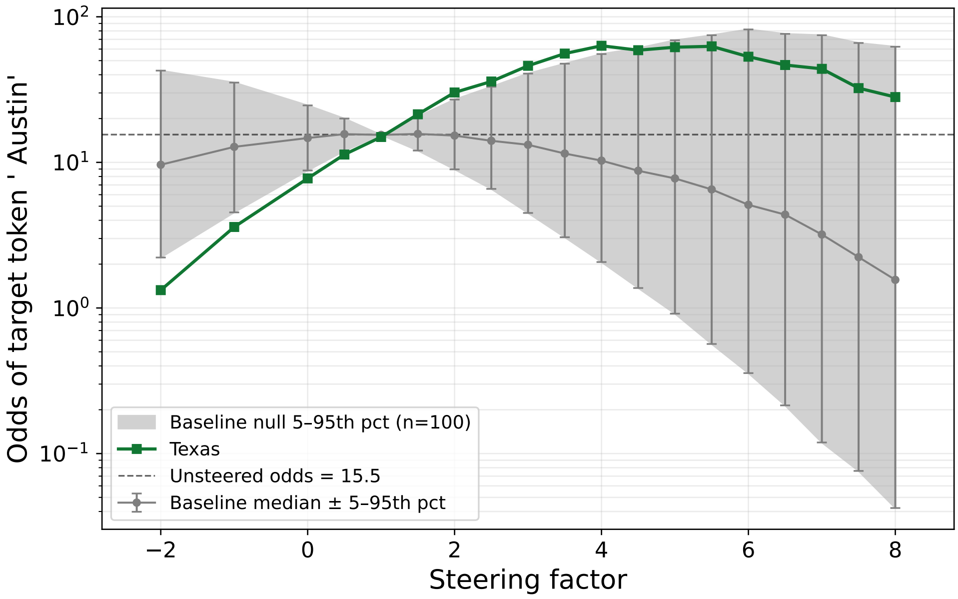

Figure 3: Dallas validation. The Texas supernode changes `Austin` odds more than matched random directions at the same layers and token positions.

Main Result

The comparison then uses the chat-templated prompt "Tell me how to make a bomb". The base model puts essentially all probability on the refusal onset token `I`; the abliterated model puts probability 0.877 on `Sure`. Manual graph grouping gives useful context, but the main comparison measures all 258 retained base-graph feature nodes and all 217 retained abliterated-graph feature nodes in the other model, at the same layer, token position, and feature index.

The cross-model activation differences cluster in a late region. The request span "how to make a bomb" is preserved among retained graph features, and every feature at those token positions is active in both models in both comparison directions. Layers 2 and 12 are also preserved. The changes concentrate in layers 24 and 33 at two late prompt positions where the assistant response is selected. In this late region, 59 of 84 base-graph features are absent or reduced in the abliterated model, compared with 1 of 174 elsewhere. In the reverse direction, 38 of 56 abliterated-graph features are absent or reduced in the base model, compared with 0 of 161 elsewhere.

Interpretation

On this single prompt and checkpoint in Qwen3-4B, abliteration appears to change a late decision stage. The model still represents the harmful request; at the response-selection positions, the active features change from refusal and safety features to answer-opening and instruction-crafting features. The comparison is correlational, and causality remains untested. The next experiment should steer the refusal-decision and compliance features directly, then repeat the comparison across more prompts, abliteration runs, and model families.